Data management plan (DMP)

Water Quality Analysis Using National Lake Monitoring Data

LAKEWATER-2026

Version	Effective date	Description of document/changes
1.0	29/05/2026	Final DMP – WP4

FWF Data Management Plan (DMP)

This DMP follows the official FWF template. When revising this DMP, please check the guidance from FWF for more details: FWF DMP - Guidance and Template

I General Information	
I.1 Administrative information	**PI:** Felix Antony Kodiyan Paul **FWF project number:** N/A – course project **Internal Project ID:** LAKEWATER-2026 **DMP version:** 1.0, 29.05.2026 **Contributors:** Ashwin Varughese Mathew, e12538359@student.tuwien.ac.at, ORCID: 0009-0002-1797-1869, TU Wien, Researcher Felix Antony Kodiyan Paul, e12436485@student.tuwien.ac.at, ORCID: 0009-0001-2323-0517, TU Wien, Researcher Anusree Rajeevan , e12534814@student.tuwien.ac.at, ORCID: 0009-0006-9044-3017, TU Wien, Researcher Joel Varghese Johnson, e12534758@student.tuwien.ac.at, ORCID: 0009-0007-0154-9707, TU Wien, Researcher
I.2 Data management responsibilities and resources	**Person responsible for data management and DMP:** Felix Antony Kodiyan Paul, e12436485@student.tuwien.ac.at, ORCID: 0009-0001-2323-0517, TU Wien **Co-ordination of data management responsibilities across partners:** Felix Antony Kodiyan Paul, e12436485@student.tuwien.ac.at, ORCID: 0009-0001-2323-0517, TU Wien **Resources costed in for RDM:** There are no costs dedicated to data management and ensuring that data will be FAIR.
II Data Characteristics	
II.1 Data description and collection or re-use of existing data	**Produced datasets:**

dataset ID	title	type	format	estimated volume	contains sensitive data	description
P1	Water Quality Classification Model	Raw data	pkl	100 - 1000 MB	no	Random Forest Classifier trained to predict water quality class (Good/Moderate/Poor) of Lithuanian lakes based on 11 physicochemical measurements. Achieves 91.9% accuracy. Serialised as water_quality_model.pkl.
P2	Generated Outputs – Water Quality Classification	Scientific and statistical data formats	CSV, PNG	100 - 1000 MB	no	Predictions, confusion matrix, label encoder and evaluation figures generated by the Random Forest classifier on the Lithuanian lakes test set.

Reused datasets:

dataset ID	title	source	rights (e.g. license)	contains sensitive data	description
R1	National Lakes Monitoring Data	https://test.dbrepo.tuwien.ac.at/database/13457a52-37f9-48d4-a078-6865e8d35981/table		no	Physicochemical measurements of Lithuanian lakes published by Aplinkos apsaugos agentūra via the European Open Data Portal. Accessed via DBRepo REST API under CC BY 4.0.

Methods and software used for data generation and reuse

New data is generated by training a Random Forest Classifier on the Lithuanian lakes dataset using Python and scikit-learn. Outputs include a trained model, label encoder, confusion matrix, and predictions CSV. All outputs are deposited on TUWRD and GitHub.

<table>
<tr><th colspan="2">III Documentation and Data Quality</th></tr>
<tr><td>III.1 Metadata and documentation</td><td>Data organisation, metadata, and documentation:

Data is structured following a 3NF relational schema in DBRepo. Code and outputs are versioned using Git on GitHub with semantic versioning (v1.0, v2.0, v3.0). Releases are archived on Zenodo (DOI: 10.5281/zenodo.20444106). File naming follows consistent conventions documented in the README.

Metadata is provided in multiple standard formats: RO-Crate (ro-crate-metadata.json), CodeMeta (codemeta.json), FAIR4ML (fair4ml.json), Croissant (croissant.json), and CITATION.cff. All deposits on TUWRD include full DataCite metadata with ORCIDs, licences, and related identifiers. This will help others to identify, discover and reuse our data.

Additionally, we will provide common metadata such as title, description, or keywords when publishing data in open access repositories. In such a case, we will follow the default template provided by the repository, such as Data Cite Metadata or Dublin Core.

As far as possible, we will use controlled vocabularies for our data to allow inter-disciplinary interoperability and machine-actionability.

Full reproduction instructions are provided in README.md. A model card (docs/model-card.md) documents the ML model. Jupyter notebooks with inline comments document all data processing steps. All outputs are deposited on TUWRD with DOIs for long-term access.</td></tr>
<tr><td>III.2 Data quality control</td><td>Data quality control:

The following data quality checks will be done: standardised data capture, data entry validation and representation with controlled vocabularies.</td></tr>
<tr><th colspan="2">IV Data Storage, Sharing, and Long-Term Preservation</th></tr>
<tr><td>IV.1 Data storage and backup during the research process</td><td>Storage and backup facilities:

For the duration of the project, storage and backup of data will be ensured by Felix Antony Kodiyan Paul (acting as the person responsible for data management and DMP) in cooperation with the system operator. The data will not be stored on the servers of TU Wien.
P1 (Water Quality Classification Model), P2 (Generated Outputs – Water Quality Classification), R1 (National Lakes Monitoring Data) will be stored on Github. External storage will be used because project uses github for version-controlled code storage and zenodo for archival, which are standard fair-compliant repositories for open research software.

Data security and protection of sensitive data:

We pay strict attention to compliance with the relevant institutional and national data protection policies. At this stage, it is not foreseen to process any sensitive data in the project. If this changes, advice will be sought from the data protection specialist at TU Wien, and the DMP will be updated.

Access to data during research:</td></tr>
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	<table><tr><th>dataset ID</th><th>selected project members</th><th>all other project members</th><th>the public</th></tr><tr><td>P1</td><td>writing</td><td>reading only</td><td>no access</td></tr><tr><td>P2</td><td>writing</td><td>reading only</td><td>no access</td></tr><tr><td>R1</td><td>writing</td><td>reading only</td><td>no access</td></tr></table> All incidents will be handled individually by an incident response team that is maintaining the affected service.
IV.2 Data sharing and long-term preservation	**Data publication and access conditions:** As far as possible, obtained datasets will be published in repositories. Details on access conditions, reuse licenses, reasons for restrictions, etc. are collected in the table below. <table><tr><th>dataset ID</th><th>access conditions</th><th>estimated publication date</th><th>location for publication (repository)</th><th>PID</th><th>license</th></tr><tr><td>P1</td><td>Open</td><td>2026-05-30</td><td>TU Wien Research Data</td><td>DOI</td><td>MIT</td></tr><tr><td>P2</td><td>Open</td><td>2026-05-30</td><td>TU Wien Research Data</td><td>DOI</td><td>CC-BY-4.0</td></tr></table> Repository description: TU Wien Research Data is an institutional repository of TU Wien to enable storing, sharing and publishing of digital objects, in particular research data. It facilitates the funders' requirements for open access to research data and the FAIR principles by making research output findable, accessible, interoperable, and reusable. A DOI is assigned to each dataset published in TU Wien Research Data. This service is developed by the TU Wien Center for Research Data Management and hosted by TU.it. https://researchdata.tuwien.ac.at/ Methods or software needed to access and use data: Users need Python 3.10+ with scikit-learn, pandas, and the dbrepo Python package to reuse the code and model. All dependencies are listed in requirements.txt. Data can be accessed via the DBRepo REST API or downloaded directly from TUWRD. All metadata is provided in standard formats (RO-Crate, CodeMeta, FAIR4ML) to facilitate reuse.

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<tr><td></td><td>Long-term preservation and deletion of data:
<table>
<tr><th>dataset ID</th><th>location for long-term storage</th><th>minimum retention period
(≥ 10 years)</th><th>foreseeable research uses and/or users</th></tr>
<tr><td>P1</td><td>TU Wien Research Data</td><td>10 years</td><td rowspan="2">Researchers and students in environmental science, data science, and machine learning who are interested in water quality analysis. Also useful for Lithuanian environmental agencies and other researchers working with lake monitoring data.</td></tr>
<tr><td>P2</td><td>TU Wien Research Data</td><td>10 years</td></tr>
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</td></tr>
<tr><td colspan="2">V Legal and Ethical Aspects</td></tr>
<tr><td>V.1 Legal aspects</td><td>Personal data:
At this stage, it is not foreseen to process any personal data in the project. If this changes, advice will be sought from the data protection specialist at TU Wien, and the DMP will be updated.
Intellectual property rights and rights of use:
The following individual(s) hold rights and control access to the project data: All 4 project members (Anusree Rajeevan, Joel Varghese Johnson, Felix Kodiyan Paul, Ashwin Varughese Mathew) have equal rights to control access. Data is publicly available under CC BY 4.0 and MIT licences.</td></tr>
<tr><td>V.2 Ethical aspects</td><td>Ethical issues:
No particular ethical issue is foreseen with the data to be used or produced by the project. This section will be updated if issues arise.</td></tr>
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